

Roll No.

TAS-801

B. TECH. (ME) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
AIRPORT AND AIRLINE MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Describe the contribution of Otto Lilienthal (The Glider Man) towards first successful glider designed. (CO2)

OR

- (b) Explain in detail the design configuration of the first heavier than air aircraft developed by the Wright brothers ? (CO2)

2. (a) What do you understand by Business Aviation ? (CO2)

OR

- (b) Why do companies rely on Business Aviation ? (CO2)

3. (a) List the top 8 challenges for the Aviation Industry in the Post-COVID Era. (CO1)

OR

- (b) What are significant threats to general aviation ? (CO1)

P. T. O.

(2)

4. (a) Show the organizational structure of the Airport Authority of India.

(CO1)

OR

- (b) What are the roles of IATA and ICAO ?

(CO1)

5. (a) Explain with neat sketch the two types of runways : Visual and Instrument.

(CO2)

OR

- (b) What do you understand by Touchdown zone markers and Landing aiming point ?

(CO1)

TME-812

B. TECH. (ME) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

NON-CONVENTIONAL ENERGY RESOURCES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is energy source ? Explain the characteristic of energy. What do you mean by non-conventional energy resources ? (CO1)

OR

- (b) Explain energy consumption and GDP in short. (CO1)

2. (a) What are natural resources ? Explain problems arising from exploitation of resources. (CO1)

OR

- (b) Explain energy planning. (CO1)

3. (a) Write energy balance equation for a typical flat plate collector (FPC) and hence derive the expression for temperature at the outlet of FPC. Also, write the various applications of FPC. (CO2)

P. T. O.

(2)

OR

(b) What do you mean by concentrator collector ? Explain with neat sketch. (CO2)

4. (a) What do you mean by direct radiation, diffuse radiation and global radiation ? Explain any *one* method of radiation measurement. (CO2)

OR

(b) Write and explain the various applications of solar collector. (CO2)

5. (a) Why should one go for storage of solar energy ? Explain solar energy storage system. (CO2)

(b) Define latitude, declination, hour angle, zenith angle solar altitude.

(CO2)

Roll No.

TME-814

B. TECH. (ME) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
ADVANCED WELDING TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Classify welding processes based on source of energy. (CO1)

OR

- (b) What is the principle of gas welding ? Explain in brief the process of gas welding. (CO1)

2. (a) What is the method of storing acetylene gas in steel cylinders ? (CO1)

OR

- (b) What are the different types of flames ? Indicate their applications and chemical reaction thereof. (CO1)

3. (a) Which welding process out of the following has maximum productivity: SAW, CO₂ Welding and TIG welding ? (CO2)

P. T. O.

(2)

OR

(b) Describe LASER beam welding process with neat sketch and applications. (CO2)

4. (a) What is the function of coated electrode ? Indicate the ingredients used for meeting the different functions of the coated electrodes. (CO1)

OR

(b) Describe manual arc welding process with neat sketch. (CO1)

5. (a) Compare TIG and MIG welding processes. (CO2)

OR

(b) Explain briefly position of welding and different joints used in welding process. (CO2)